0580_w24_ms_41_Q4

Question	Answer	Marks	Partial Marks	Guidance
4(a)(i)	22.5	**2**	**M1** for $\frac{11.25}{11.25+18.75+20}$ [×100] oe	2 marks for $22\frac{1}{2}$ M1 for eg 11.25/50, 9/40, 900/40, 90/4 , 45/2
4(a)(ii)	9 : 15 : 16	**2**	**M1** for 1125 : 1875 : 2000 or better	For M1 all must be integers e.g. 45 : 75 : 80
4(a)(iii)(a)	$\frac{5}{9}$ or 0.556 or 0.5555 to 0.5556	**3**	**M2** for $\frac{20\times 25}{15[\times 60]}$ oe or **M1** for 20 × 25 or for *their* distance ÷ (15 [× 60]) oe	Accept $0.\dot{5}$ or other 'recurrent' notation Mark at most accurate Eg $\frac{500}{900}$ is M2 only as in context. M2 implied by 33.3[3…] or 0.555 but 0.56 alone does not imply method 500 *their* distance only 25, or 25 ×11.25, or 25 ×18.75 oe eg $\frac{25}{900}$

0580_w24_ms_41_Q4

Question	Answer	Marks	Partial Marks	Guidance
4(a)(iii)(b)	2 h 40 mins	4	**<u>Approach 1</u>** **B3** for $\frac{8}{3}$ [h]oe or 160 [mins] or 9600[s] Or **M3** for 5000 ÷ (18.75 x 25 x 4)[h] oe or 5000 ÷ (18.75 x 25 ÷ 15)[mins] oe or 5000 ÷ ((18.75 x 25 x 4) ÷ (60 x 60))[secs] oe Or **M2** for (18.75 x 25 x 4)[m/h] oe or (18.75 x 25 ÷ 15)[m/min] oe or (18.75 x 25 x 4) ÷ (60 x 60))[m/sec] oe Or **B1** for 200 or 1 km =1000m soi After 0 scored **SC1** for time *Figs* 267 or *figs* 2666 to 2667 or *figs* 16 or *figs* 96 **<u>Approach 2</u>** **B3** for 160 [mins] Or **M3** for 15 x 5000 ÷ (18.75 x 25) [mins] oe Or **M2** for 5000 ÷ (18.75 x 25) oe Or **B1** for 200 or 1 km =1000m soi After 0 scored **SC1** for time *figs* 16	For B3 isw time conversion oe only $2\frac{2}{3}$ or 2.67 or 2.666 to 2.667 1875[m/h] 31.25[m/min] 0.5208[…][m/s], 0.521, $\frac{25}{48}$, 468.75 ÷ 900 Soi by eg 5000, 0.025, 0.01875 $15 \times \frac{32}{3}$ 5000 ÷ 468.75, $\frac{32}{3}$ Soi by eg 5000, 0.025, 0.01875 **See AG for misinterpretation of question**

0580_w24_ms_41_Q4

Question	Answer	Marks	Partial Marks	Guidance
4(a)(iv)	17.1 or 17.14 to 17.15	**3**	**M2** for $20 \times \left(\frac{100-5}{100}\right)^3$ oe or **M1** for $20 \times \left(\frac{100-5}{100}\right)^k$ where k is 2, or 4 or for $20 \times \left(\frac{100-5}{100}\right)^3$ oe seen and spoiled	Accept answer 17 only after M2 earned For M1 or M2 do not accept $20 \times (95\%)^k$ May be done in stages: 19, 18.05, 17.15. If done in stages then 3sf or better at each stage implies method eg, 19, 18.1, 17.2 M1 implied by answer 18.05 or 16.3 or 16.29 to 16.3 If done in stages do not allow M1 if they go beyond 4 stages.
4(b)	2500	**3**	**M2** for $\frac{425 \text{ to } 450}{10+0.5}$ or $\frac{450-12.5}{10 \text{ to } 11}$ or $\frac{425 \text{ to } 450}{630}$ or $\frac{450-12.5}{600 \text{ to } 660}$ or **M1** for 10.5 or 9.5 or 437.5 or 462.5 or 630[s] or 570[s]	Mark at most accurate. Eg 2499.9 = 2500 scores maximum of M2 (candidates convert to m/s and lose accuracy) M2 implied by 41.7 or 41.66 to 41.67 or $\frac{125}{3}$ 10 + 0.5 oe or 10 – 0.5 oe or 450 – 12.5 oe or 450 + 12.5 oe